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Applications of Thales' theorem

The theorem inside a triangle: proportional segments, the line through a midpoint, and the measurements you cannot reach with a tape.

LESSON 13–14 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 9–10 | STAGE 9 · 13.4

LESSON CLOCK

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Apply Thales' theorem inside a triangle to find unknown lengths.
- Prove that a line through the midpoint of one side, parallel to a second, bisects the third.
- Divide a segment in a given ratio by construction.
- Solve practical measuring problems using proportional segments.

TEXTBOOK

National	Темы 9–10, pp. 23–28
Cambridge	Learner's Book pp. 293–299
Workbook	Workbook 13.4

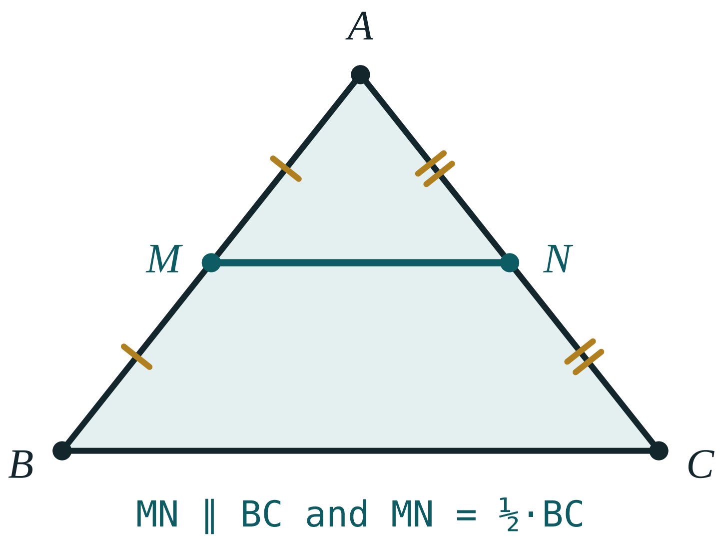
01 Explanation

Thales inside a triangle

THE FORM YOU WILL USE MOST

If a line parallel to BC cuts AB at M and AC at N, then

$$\frac{AM}{MB} = \frac{AN}{NC} \quad \text{and} \quad \frac{AM}{AB} = \frac{AN}{AC}$$



When M and N are the midpoints, the ratio is 1 : 1 — the special case that gives the midline.

Two different ratios, and mixing them up is the commonest error in this topic. Decide first whether you are comparing a **part with a part** ($AM : MB$) or a **part with the whole** ($AM : AB$), and stay with that choice on both sides.

The midpoint corollary

COROLLARY

A line drawn through the midpoint of one side of a triangle, parallel to a second side, bisects the third side.

Proof. With $AM = MB$ the ratio $\frac{\mathrm{AM}}{\mathrm{MB}}$ is 1, so $\frac{\mathrm{AN}}{\mathrm{NC}}$ is also 1 and $AN = NC$. ∎

This corollary is what makes the midline theorem of the next lesson possible, so it is worth stating separately.

Dividing a segment in a given ratio

To divide AB in the ratio $2 : 3$:

1. Draw a ray from A and step off $2 + 3 = 5$ equal segments.
2. Join the 5th point to B.
3. Through the 2nd point draw a parallel to that line.
4. It meets AB at the point dividing it in the ratio $2 : 3$.

The same construction, with n equal steps and a parallel through each, divides the segment into n equal parts.

Measuring what you cannot reach

The practical use of the theorem is measuring across an obstacle — a river, a building, a fenced field. Set up two similar configurations sharing an angle, measure the three lengths you can reach, and solve the proportion for the fourth.

WORKED PATTERN

A pole $1.5\ m$ tall casts a shadow $2\ m$ long; at the same moment a tree casts a shadow $14\ m$ long. The sun's rays are parallel, so

$$\frac{1.5}{2} = \frac{h}{14} \implies h = 10.5\ m$$

02 Worked examples

EXAMPLE 1 ***In $\triangle ABC$, $MN \parallel BC$ with $AM = 4$, $MB = 6$ and $AN = 6$. Find NC.***

1. $$\frac{AM}{MB} = \frac{AN}{NC}$$
   Part with part on both sides.

2. $$\frac{4}{6} = \frac{6}{NC}$$
   Substitute.

3. $4 \cdot NC = 36$
   Cross-multiply.

4. $NC = 9$

ANSWER

$NC = 9$

EXAMPLE 2 ***In △ABC, MN ∥ BC with AM = 3, AB = 12 and AC = 20. Find AN.***

1. $$\frac{AM}{AB} = \frac{AN}{AC}$$
   This time it is part with whole — note AB, not MB.

2. $$\frac{3}{12} = \frac{AN}{20}$$

3. $12 \cdot AN = 60$

4. $AN = 5$

ANSWER

$AN = 5$

EXAMPLE 3 **A 1.8 m man casts a 2.4 m shadow. A tower casts a 40 m shadow at the same time. Find its height.**

1. The sun’s rays are parallel, so the two right triangles have proportional sides.
   That is Thales in disguise.

2. $$\frac{1.8}{2.4} = \frac{h}{40}$$
   Height over shadow, on both sides.

3. $2.4h = 72$

4. $h = 30\ m$

ANSWER

$30\ m$

03 Interactive model

Move the spacing slider so the cuts are unequal, and check that the ratios still match on both rays.

Which proportion do I write?

> Given as **part and part** on the first side.

1. The data give

AM and

MB — two pieces of the same side.

So use part-to-part.

2. $$\frac{\mathrm{AM}}{\mathrm{MB}} = \frac{\mathrm{AN}}{\mathrm{NC}} \Longrightarrow \frac{4}{6} = \frac{6}{\mathrm{NC}}$$

3. $4 \cdot NC = 36$, so $NC = 9$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Proportion	Proporsiya	Пропорция
Cross-multiply	Krest-nakrest ko‘paytirish	Умножить крест-накрест
Midpoint	O‘rta nuqta	Середина
Scale	Masshtab	Масштаб
Shadow	Soya	Тень
Similar triangles	O‘xshash uchburchaklar	Подобные треугольники
Part to whole	Qism-butunga nisbat	Отношение части к целому
Part to part	Qism-qismga nisbat	Отношение части к части

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$MN \parallel BC$ with $AM = 2$, $MB = 3$, $AN = 4$. Then NC is:

- 6
- 5
- 8
- 2.4

$MN \parallel BC$ with $AM = 2$, $AB = 5$, $AC = 15$. Then AN is:

- 6
- 7.5
- 10
- 3

A line through the midpoint of AB parallel to BC meets AC at N. Then N is:

- a third of the way along AC
- the midpoint of AC
- not determined
- the same point as C

A 2 m pole casts a 3 m shadow. A 24 m shadow belongs to an object of height:

- *16 m*
- *36 m*
- *12 m*
- *48 m*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $MN \parallel BC$, $AM = 2$, $MB = 3$, $AN = 4$. Find NC.

   ANSWER 6

2. $MN \parallel BC$, $AM = 5$, $MB = 5$, $AN = 7$. Find NC.

   ANSWER 7

3. $MN \parallel BC$, $AM = 3$, $AB = 9$, $AC = 12$. Find AN.

   ANSWER 4

4. *A 2 m pole casts a 3 m shadow; a tree casts 24 m. Find its height.*

   ANSWER 16 m

5. M is the midpoint of AB and $MN \parallel BC$. What is N?

   ANSWER The midpoint of AC.

6. $MN \parallel BC$ and $AM : MB = 1 : 2$. Find $AN : NC$.

   ANSWER $1 : 2$

7. $MN \parallel BC$, $AM = 4$, $AN = 4$, $NC = 8$. Find MB.

   ANSWER 8

Medium · the standard question

7 PROBLEMS

1. $MN \parallel BC$, $AM = 4$, $MB = 6$, $AN = 6$. Find NC.

   ANSWER 9

2. $MN \parallel BC$, $AM = 3$, $AB = 12$, $AC = 20$. Find AN.

   ANSWER 5

3. *A 1.8 m man casts a 2.4 m shadow; a tower casts 40 m. Find its height.*

   ANSWER 30 m

4. $MN \parallel BC$, $AN : NC = 3 : 5$ and $AB = 24$. Find AM.

   ANSWER 9

5. *Describe how to divide a segment in the ratio 2 : 3.*

   ANSWER 5 equal steps on a ray, join the 5th to B, draw the parallel through the 2nd.

6. $MN \parallel BC$, $AM = x$, $MB = 8$, $AN = 6$, $NC = 12$. Find x.

   ANSWER $x = 4$

7. $MN \parallel BC$ and MN cuts off a triangle with $AM : AB = 2 : 7$. Find $AN : AC$.

   ANSWER $2 : 7$

Hard · stretch and reason

7 PROBLEMS

1. *Prove that a line through the midpoint of AB parallel to BC bisects AC.*

   ANSWER $\frac{AM}{MB} = 1$ forces $\frac{AN}{NC} = 1$ by Thales, so $AN = NC$.

2. *In $\triangle ABC$, $MN \parallel BC$, $AM = x$, $MB = x + 3$, $AN = 4$, $NC = 8$. Find x.*

   ANSWER $\frac{x}{x+3} = \frac{4}{8} \implies 2x = x + 3 \implies x = 3$

3. *A river is crossed by sighting: two parallel lines cut a base line into 12 m and 18 m, and one side of the river measures 20 m. Find the far distance.*

   ANSWER $\frac{12}{18} = \frac{20}{d} \implies d = 30\ m$

4. *$MN \parallel BC$ with $AM : MB = 3 : 4$. The perimeter of $\triangle AMN$ is 21. Find the perimeter of $\triangle ABC$.*

   ANSWER Sides scale by $\frac{3}{7}$, so the perimeter of $\triangle ABC$ is 49.

5. *Divide a segment into 7 equal parts and explain why no measurement of its length is needed.*

   ANSWER Seven equal compass steps on a ray plus parallels transfer the equality by Thales; the length of AB is never used.

6. *In trapezium ABCD ($AD \parallel BC$), the diagonals meet at O. Prove* $\frac{AO}{OC} = \frac{AD}{BC}$.

   ANSWER The parallels AD and BC cut the transversals AC and BD proportionally.

7. *Three parallels cut one transversal into 4 cm and 6 cm and another into pieces whose sum is 20 cm. Find them.*

   ANSWER Ratio $2 : 3$ of 20 gives $8\ cm$ and $12\ cm$.

07 Homework

Homework — 6 problems

Geometry 8, Темы 9–10, pp. 23–28. Write the proportion before you substitute.

1. *$MN \parallel BC$, $AM = 6$, $MB = 9$, $AN = 8$. Find NC.*
2. *$MN \parallel BC$, $AM = 5$, $AB = 15$, $AC = 21$. Find AN.*
3. *A 1.6 m person casts a 2 m shadow; a flagpole casts 15 m. Find its height.*
4. *$MN \parallel BC$ and $AM : MB = 2 : 5$. Find $AN : AC$.*
5. *Divide a 10 cm segment in the ratio 3 : 2 by construction.*
6. *In $\triangle ABC$, $MN \parallel BC$ and the perimeter of $\triangle AMN$ is half that of $\triangle ABC$. Find $AM : MB$.*